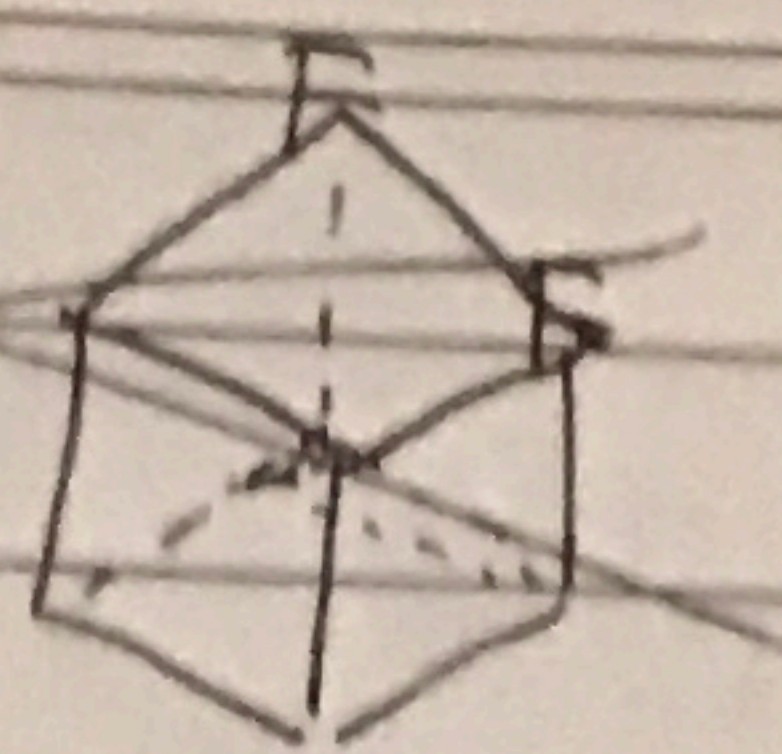
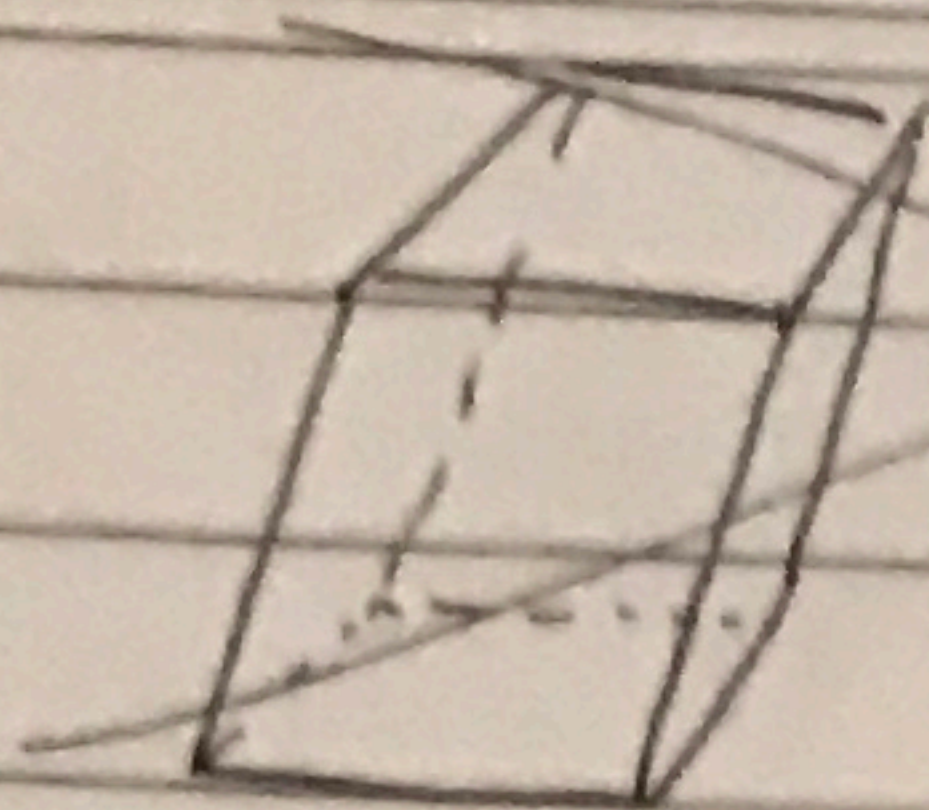
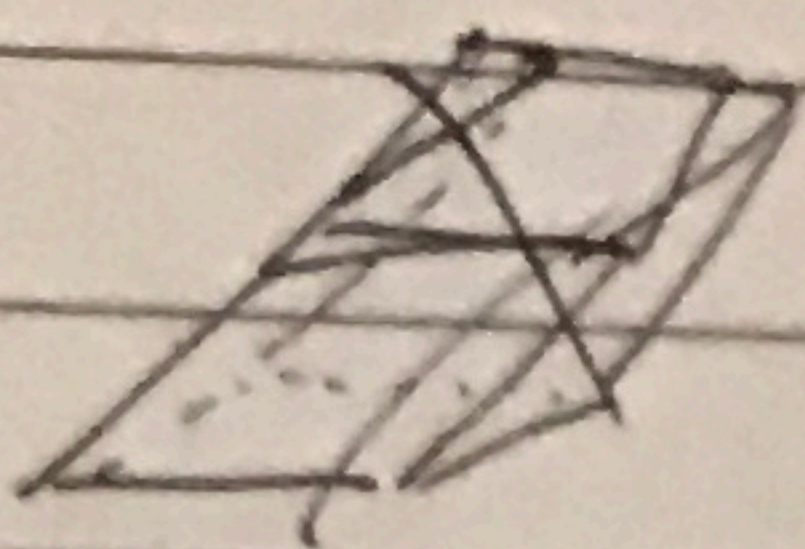
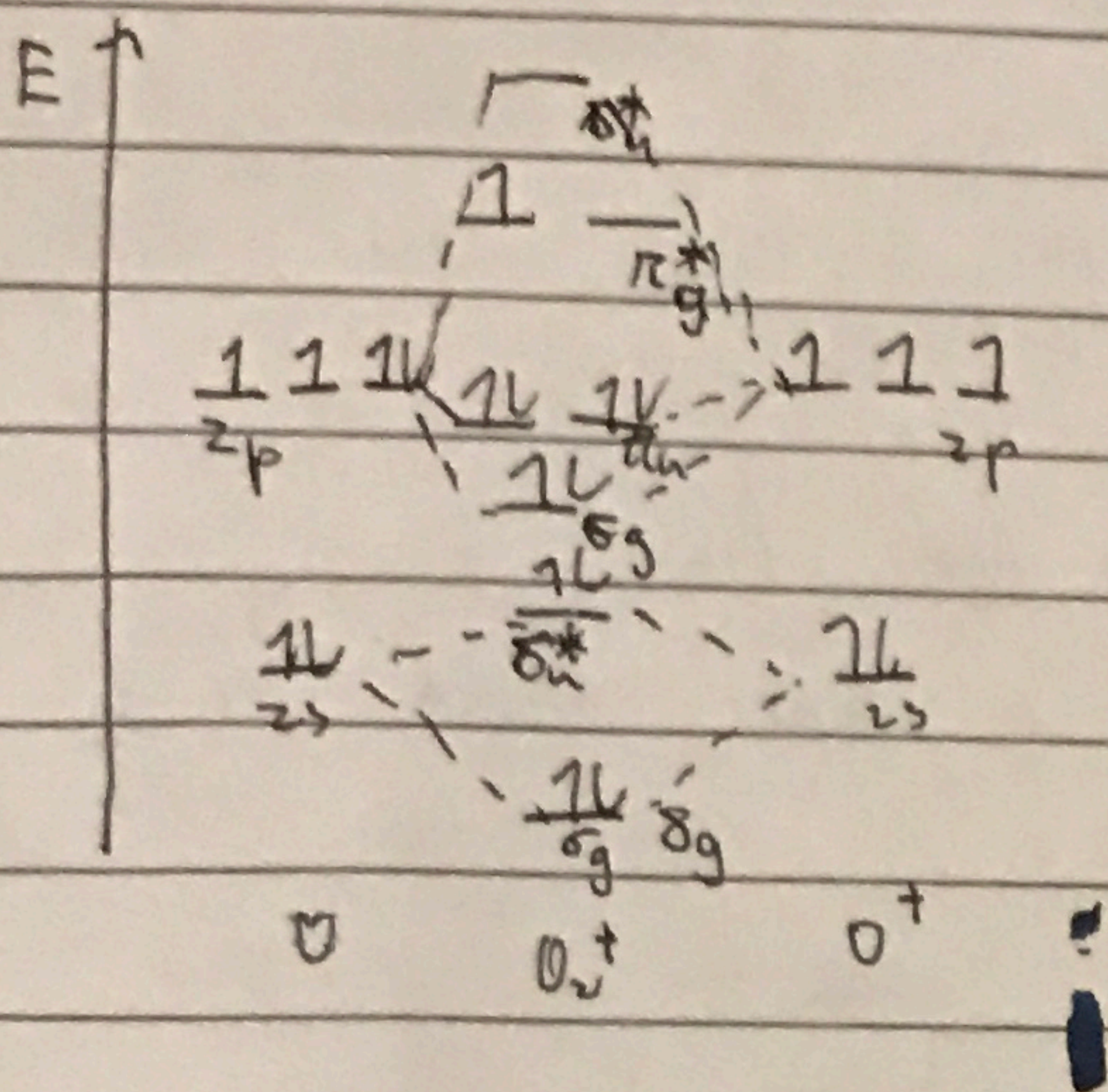


(a) n

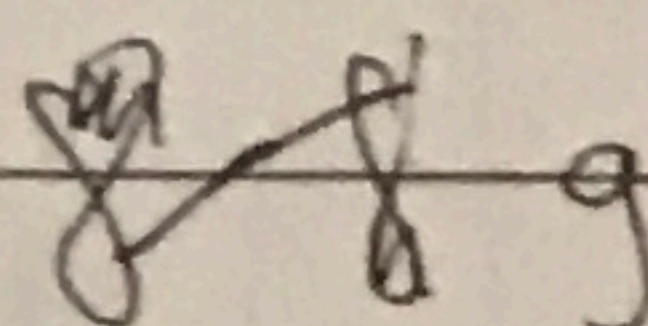
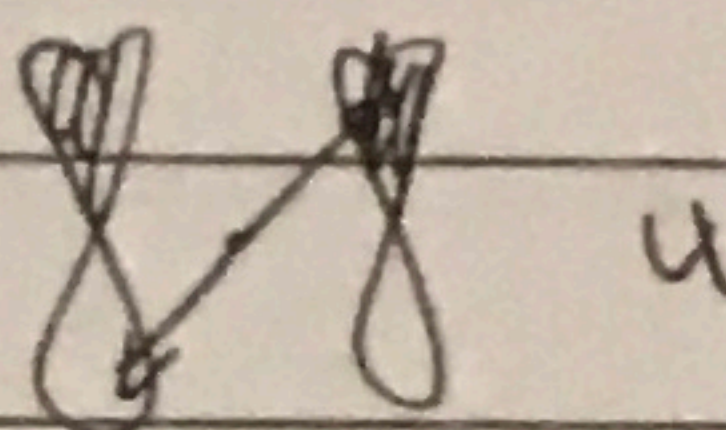
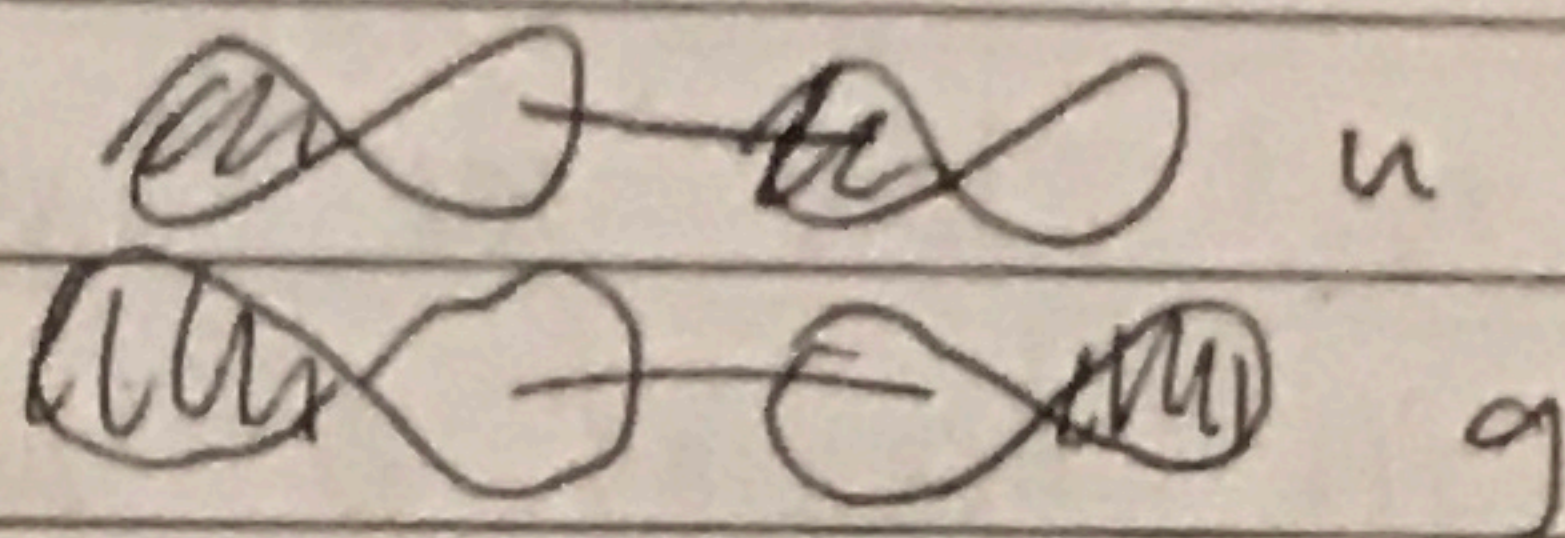
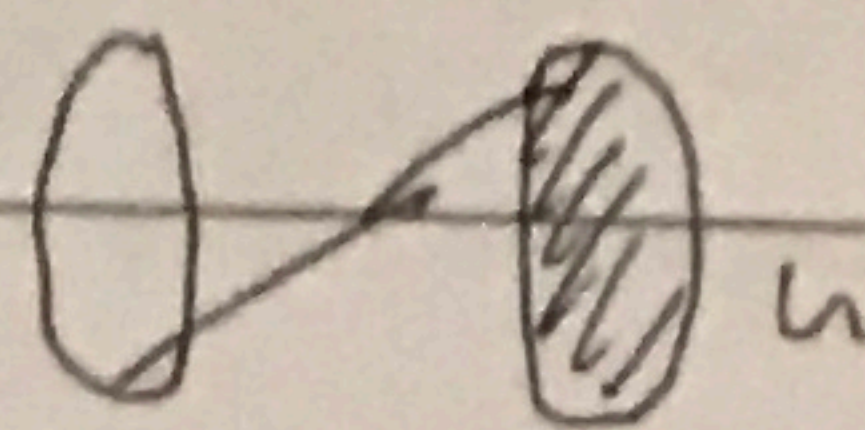
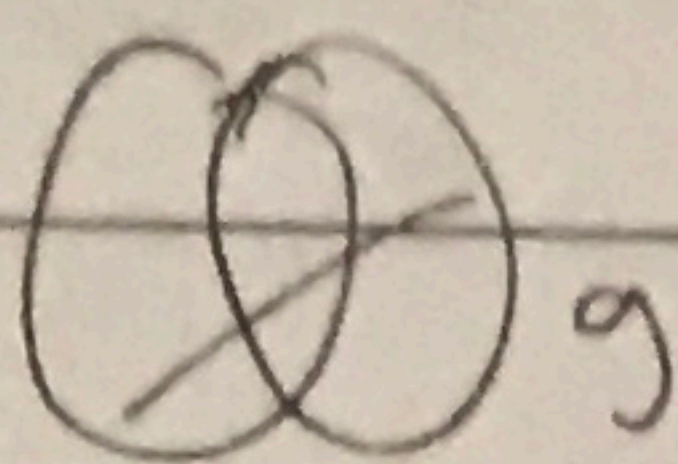
O_2 PtF_6



O_2^+ and PtF_6^-



paramagnetic,
bond length shorter
than O_2 as there is
less electron in the
antibonding MOs



KO_2

K^+

O_2^-

extra electron in
antibonding MOs weakened
bond, paramagnetic

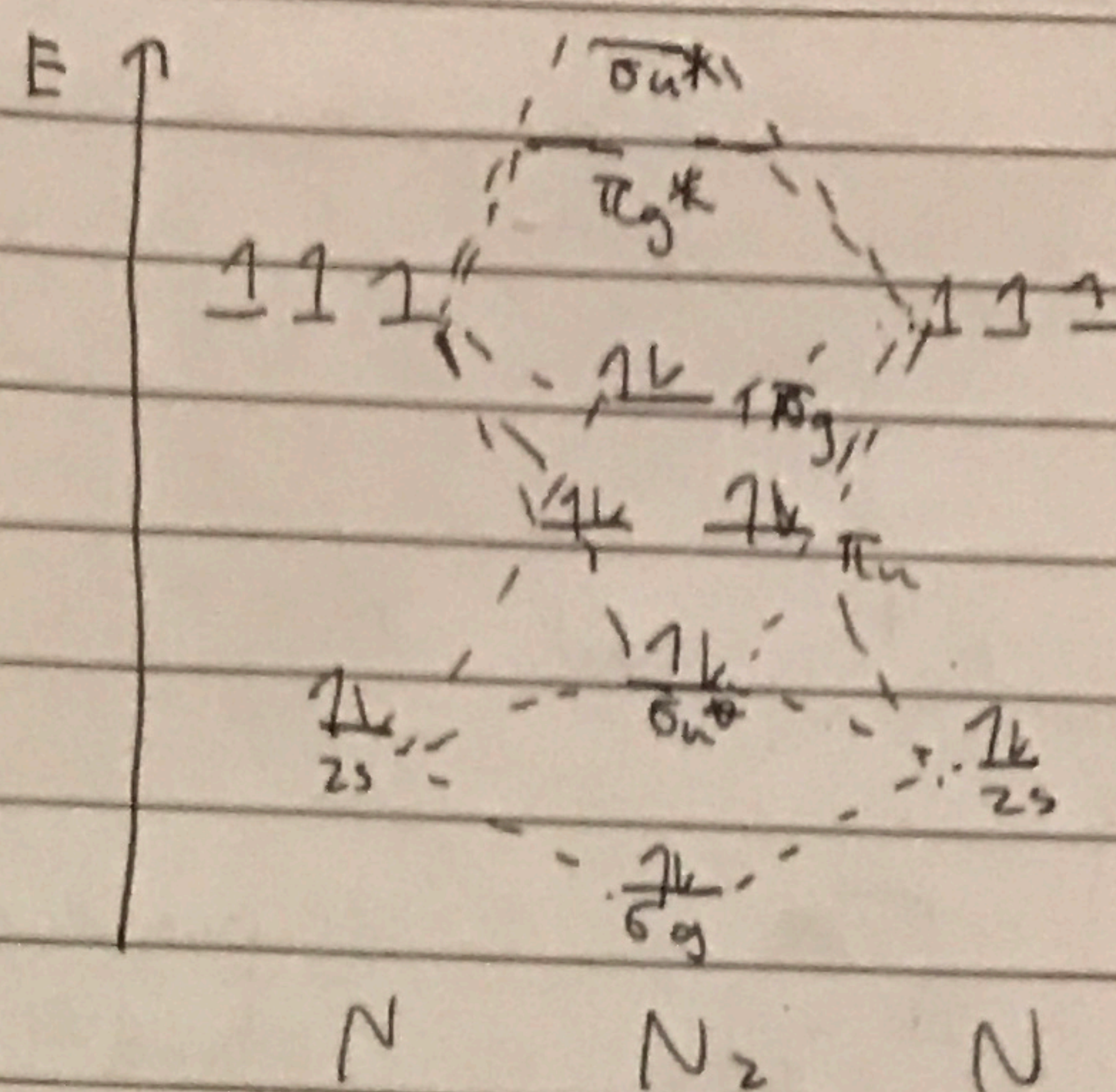
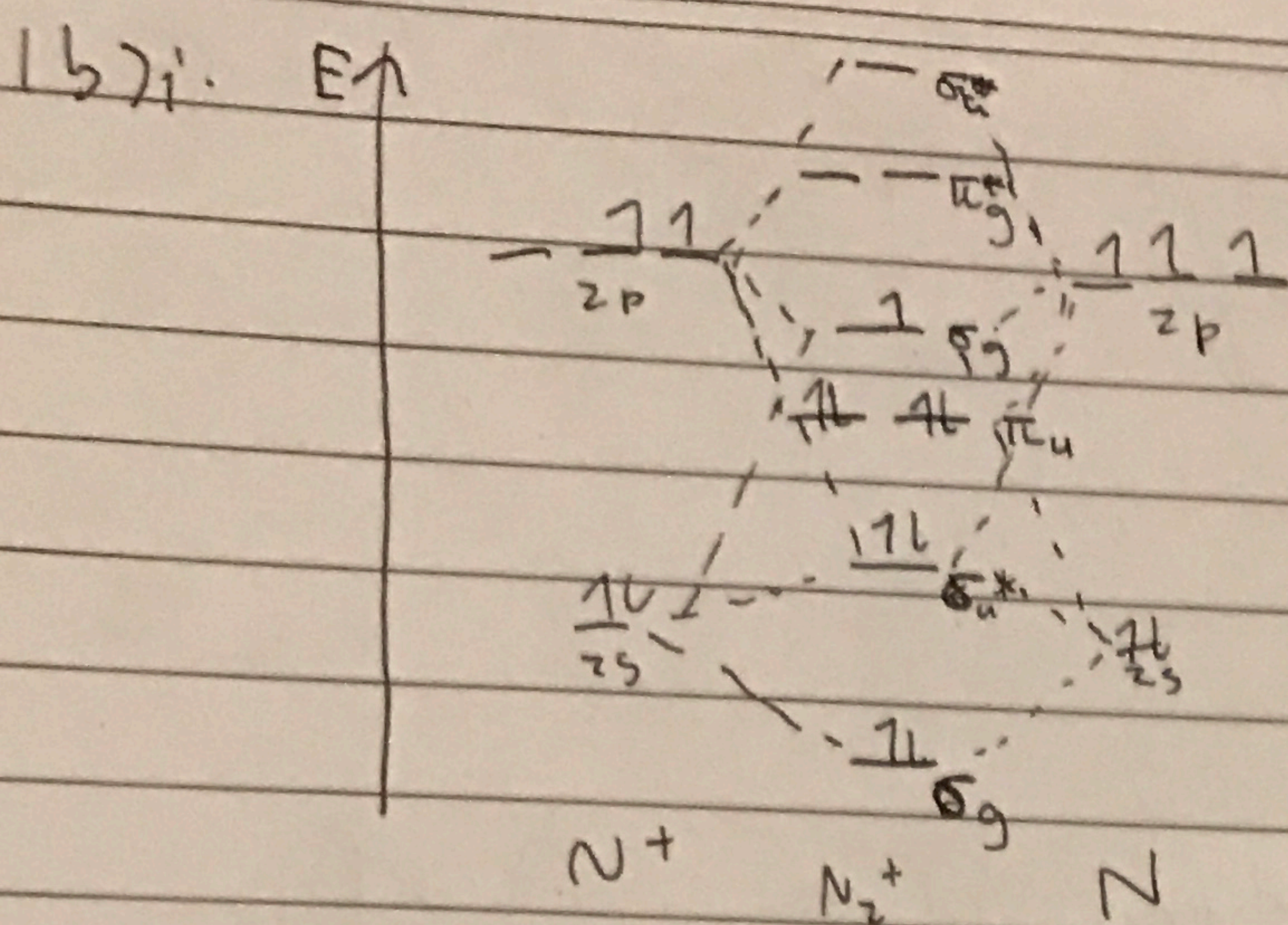
O_2 paramagnetic

Na_2O_2

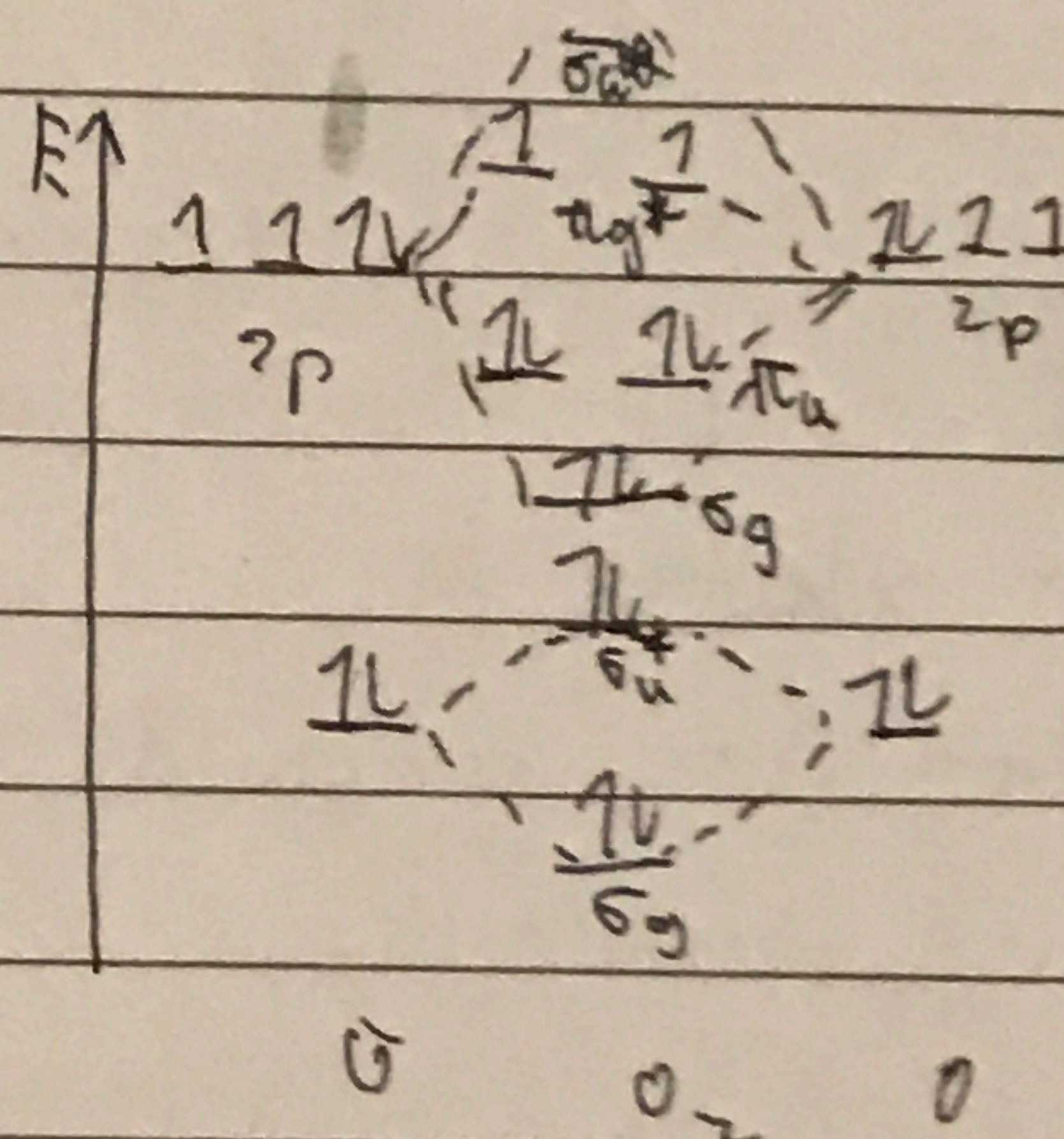
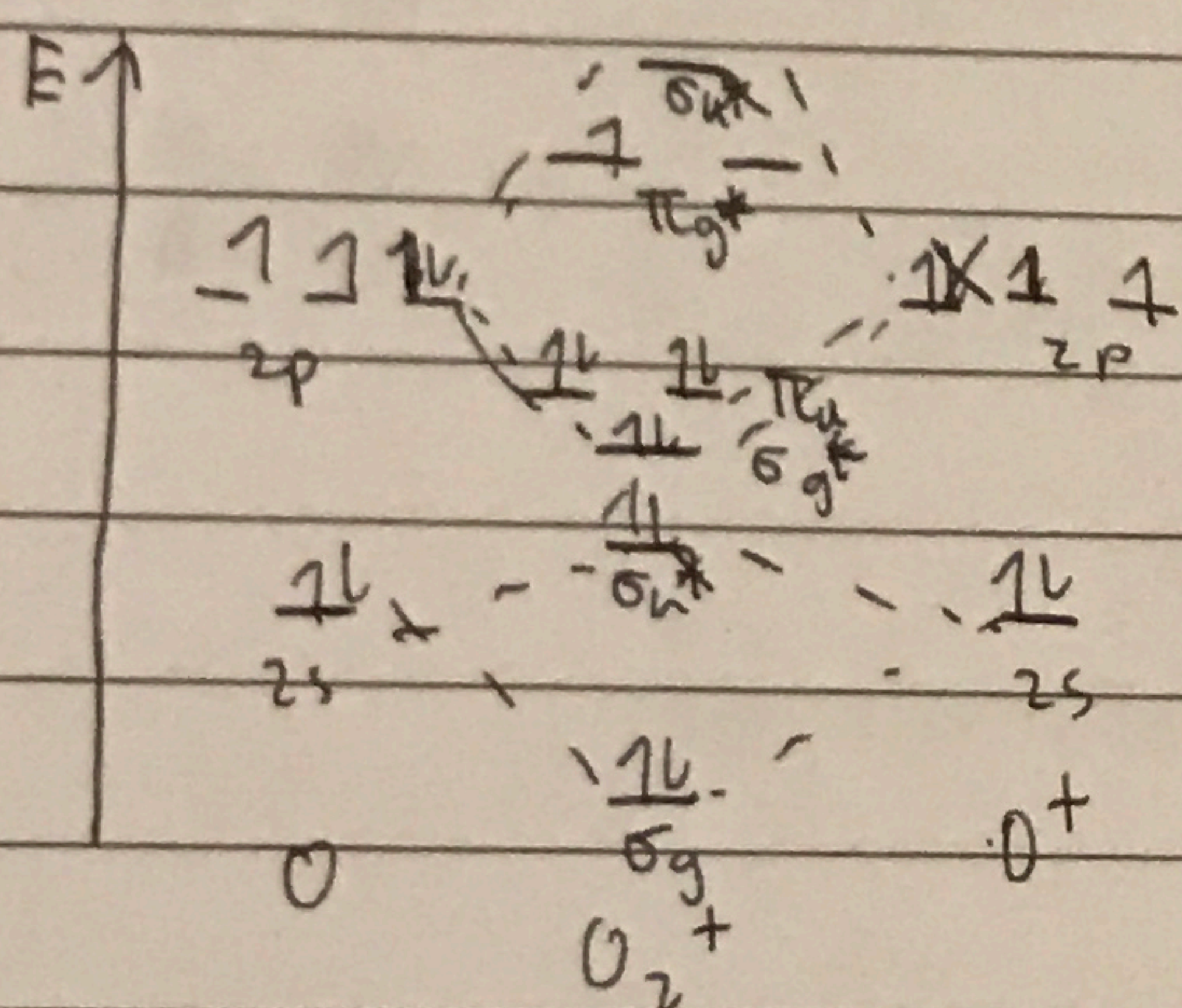
Na^+

O_2^{2-}

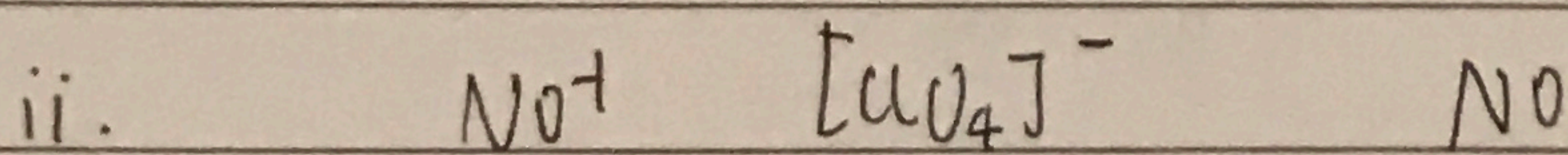
2 extra electrons in
antibonding MOs, weakened
bond, diamagnetic



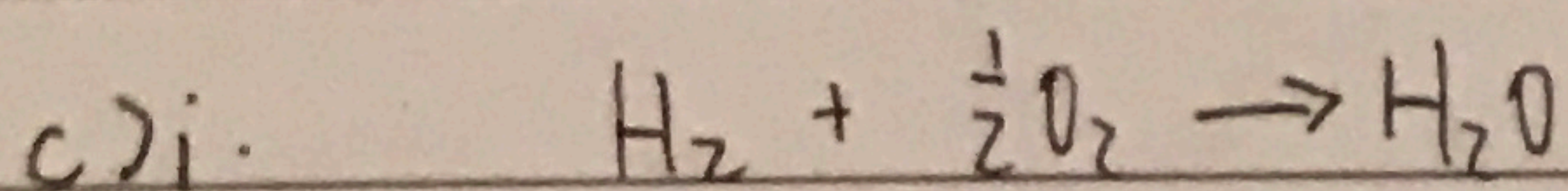
N_2^+ has one less electron in its bonding MOs than N_2 , so its bonding is weaker, therefore it has a longer bond length.



O_2 has one more electron in its antibonding MOs than O_2^+ , so its bonding is weaker, thus, it has a longer bond length.



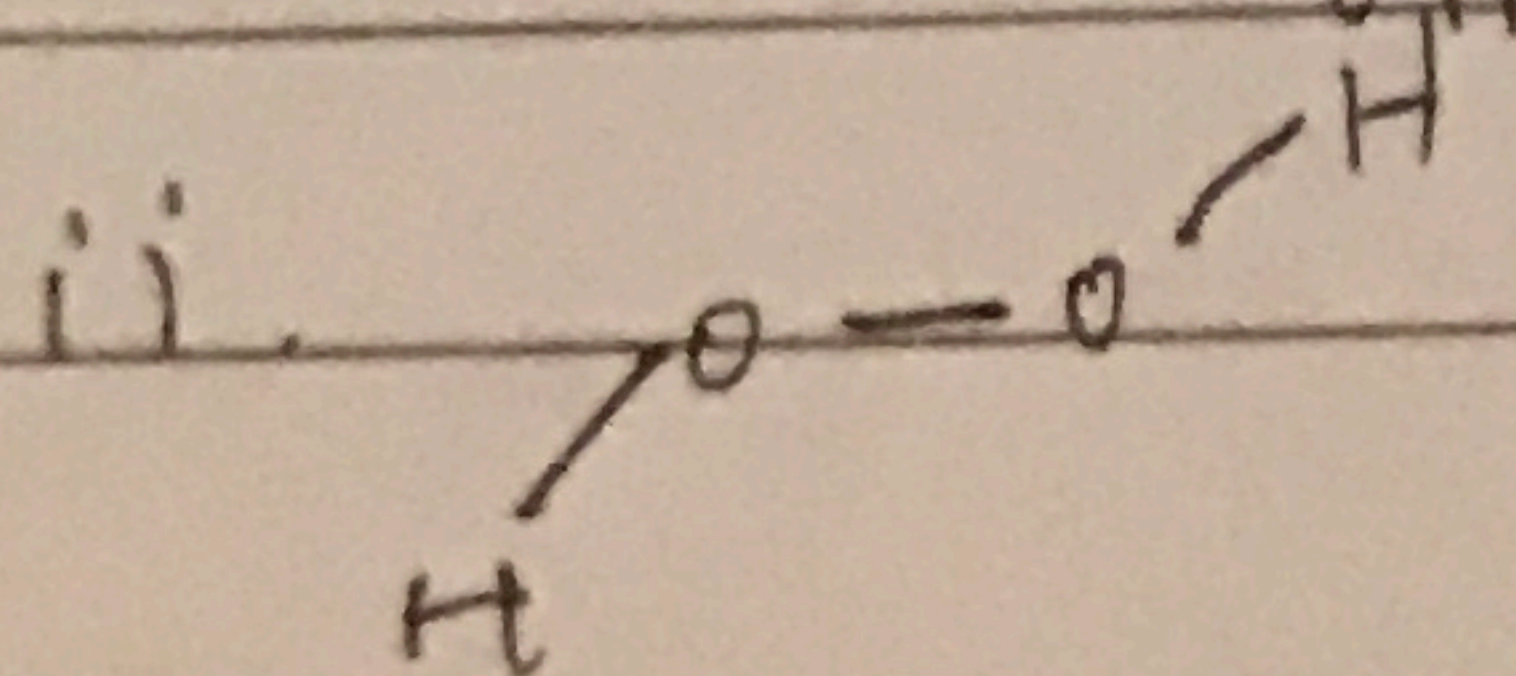
there is one more electron in the antibonding MOs of NO compared to the NO^+ in $NO[NO_4]^-$, so the bond is weaker and thus longer.



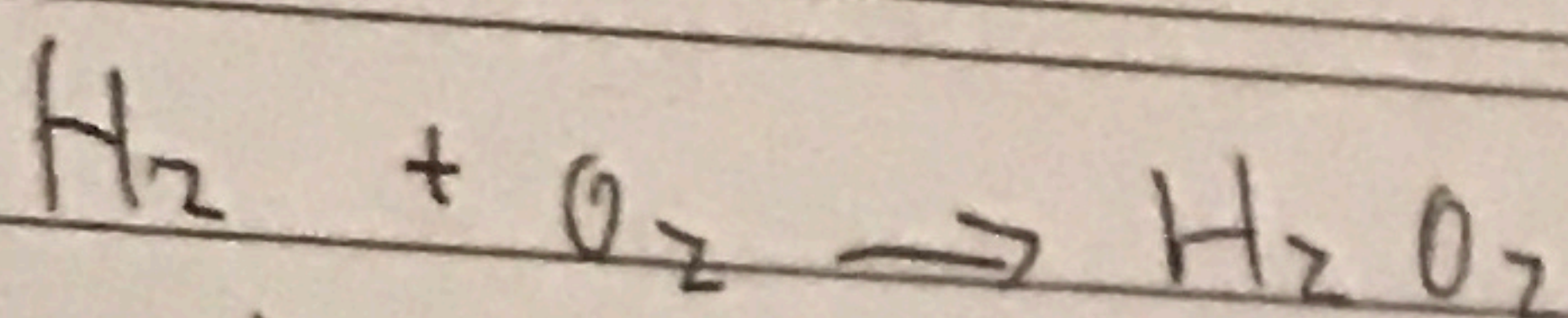
1 mol $H-H$ broken, $\frac{1}{2}$ mol $O-O$ broken, 2 mol $H-O$ formed

$$-242 = 436 + \frac{498}{2} - 2E_{O-H}$$

$$E_{O-H} = 463.5 \text{ kJ/mol}$$

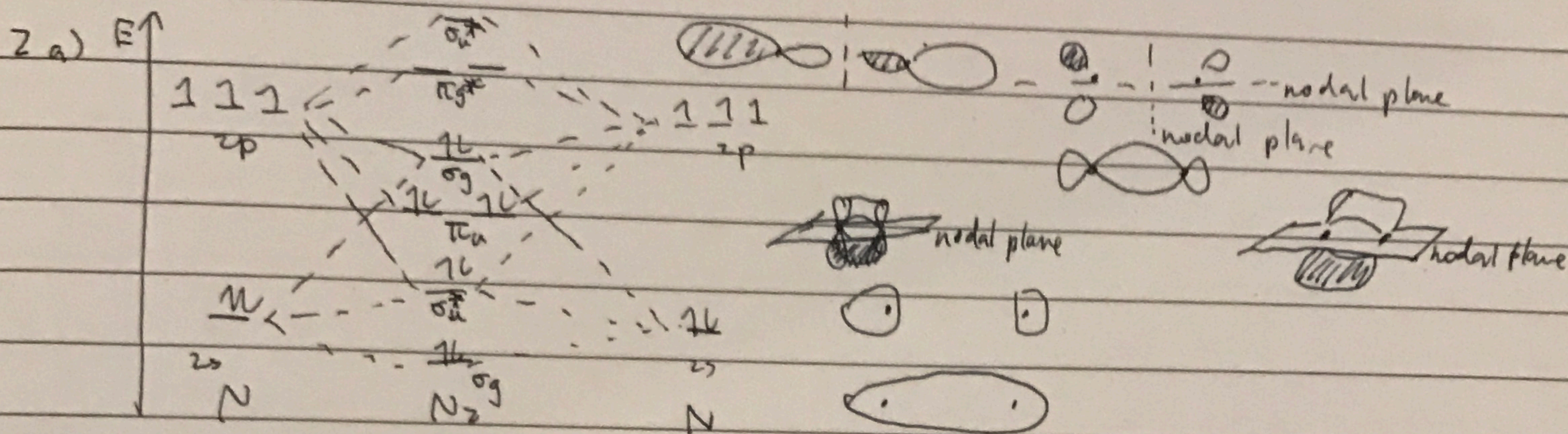


c) ii.



$$-136 = 436 + 498 - 436 - E_{0-0}$$

$$E_{0-0} = 634 \text{ kJ/mol}$$



b) bond order = 3

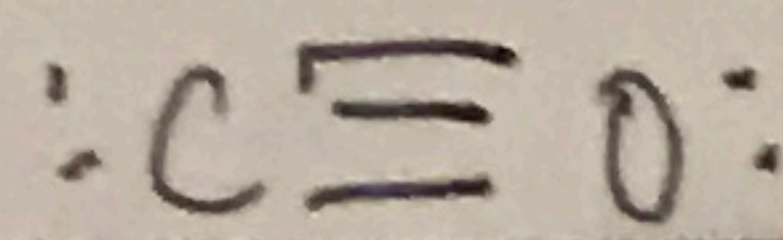
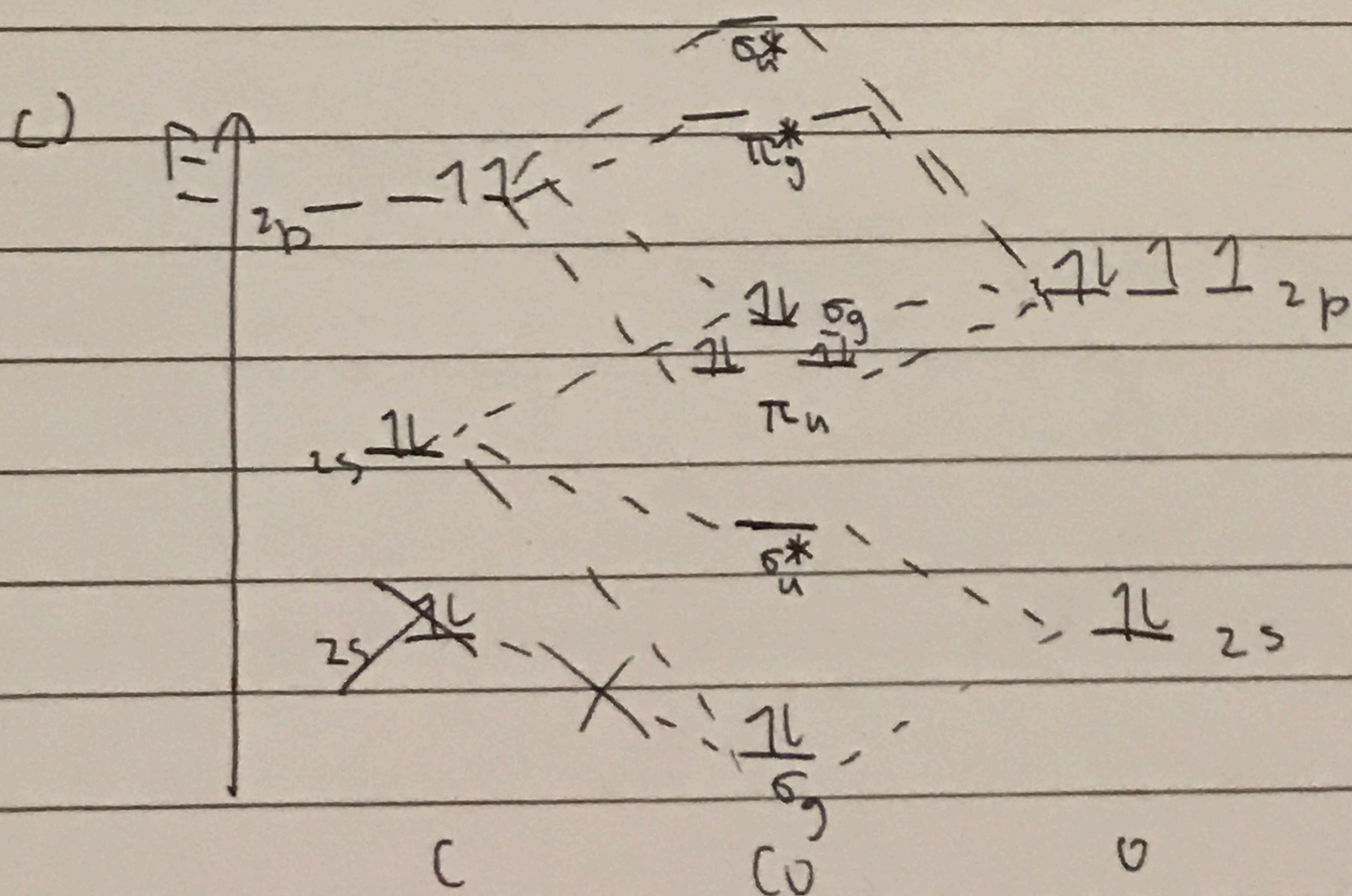
~~strong bonding~~

strong bonding — inert usually

short bond length

diamagnetic

IR inactive



oxygen is more electronegative